Exercise C - II

1. If $\cos(x-y)$, $\cos x$ and $\cos(x+y)$ are in H.P., then $\cos x \sec\left(\frac{y}{2}\right) =$ _____.

यदि $\cos(x-y)$, $\cos x$ एवं $\cos(x+y)$ हरात्मक श्रेढी में हो तो $\cos x \sec\left(\frac{y}{2}\right) =$ _____

Ans. $\pm\sqrt{2}$

Sol. $\cos(x-y), \cos x, \cos(x+y) \rightarrow$ H.P.

$$\Rightarrow \cos x = \frac{2\cos(x-y)\cos(x+y)}{\cos(x-y)+\cos(x+y)}$$

$$\Rightarrow \cos x = \frac{\cos^2 x - \sin^2 y}{\cos x \cdot \cos y}$$

$\Rightarrow \cos^2 x \cos y = \cos^2 x - \sin^2 y$

$\Rightarrow \sin^2 y = \cos^2 x (1 - \cos y)$

$$\Rightarrow 4\sin^2\frac{y}{2}\cos^2\frac{y}{2} = \cos^2 x \cdot 2 \cdot \sin^2\frac{y}{2}$$

$$\Rightarrow \left[\cos x \sec\left(\frac{y}{2}\right)\right]^2 = 2$$

$$\Rightarrow \cos x \sec\left(\frac{y}{2}\right) = \pm\sqrt{2}$$

2. If in ΔABC, $\sin A \sin B \sin C + \cos A \cos B = 1$, then the value of $\sin C =$ _____.

किसी Δ ABC में $\sin A \ \sin B \ \sin C + \cos A \ \cos B = 1$ तब $\sin C =$ _____

Ans. 1

Sol. $\cos A \ \cos B + \sin A \sin B \sin C = 1$

$\Rightarrow 2 \cos A \cos B + 2 \sin A \sin B \sin C = 2$

$\Rightarrow 2 \cos A \cos B + 2 \sin A \sin B \sin C = \sin^2 A + \cos^2 A + \sin^2 B + \cos^2 B$

$\Rightarrow \cos^2 A + \cos^2 B - 2 \cos A \cos B + \sin^2 A + \sin^2 B - 2 \sin A \sin B + 2 \sin A \sin B - 2 \sin B \sin A \sin C = 0$

$\Rightarrow (\cos A - \cos B)^2 + (\sin A - \sin B)^2 + 2\sin A \sin B (1 - \sin C) = 0$

$\because$ In ΔABC; it is true if

$\cos A = \cos B$ & $\sin A = \sin B$ & $\sin C = 1$

$\Rightarrow A = B$ & $\sin C = 1$

3. Given that for a, b, c, d ∈ R, if a sec(200°) – c tan(200°) = d and b sec(200°) + d tan(200°) = c, then find the value of $\left(\frac{a^2+b^2+c^2+d^2}{bd-ac}\right)\sin 20°$.

यदि a sec(200°) – c tan(200°) = d तथा b sec(200°) +d tan(200°) = c जहां a, b, c तथा d ∈ R वास्तविक संख्याऐं है, तो $\left(\frac{a^2+b^2+c^2+d^2}{bd-ac}\right)\sin 20°$ का मान बताओं।

Ans. 2

Sol. Given

a sec (200º) – c tan (200º) = d

& b sec (200º) + d tan (200º) = c

$\Rightarrow$ a sec (180º + 20º) – c tan (180º + 20º) = d

& b sec (180º + 20º) + d tan (180º + 20º) = c

$\Rightarrow$ –a sec 20º – c tan 20º = d ...(i)

& –b sec 20º + d tan 20º = c ...(ii)

By (i) · d + (ii) · c; we get

$$\sec 20^\circ = \frac{d^2+c^2}{-ad-bc} \quad ...(iii)$$

By (i) · b – (ii) · a; we get

$$\tan 20^\circ = \frac{bd-ac}{-bc-ad} \quad ...(iv)$$

By (iii) & (iv)

$$\sec 20^\circ = \frac{d^2+c^2}{(bd-ac)}\cdot\tan 20^\circ$$

$$\Rightarrow \left(\frac{d^2+c^2}{bd-ac}\right)\sin 20^\circ = 1 \quad ...(v)$$

Now, by (i) · a + (ii) · b; we get

– ($a^2 + b^2$) sec 20º + tan 20º (bd – ac) = ad + bc

$\Rightarrow$ ($a^2 + b^2$) sec20º = tan 20º (bd – ac) – (ad + bc)

$$\Rightarrow \left(\frac{a^2+b^2}{bd-ac}\right)\sec 20^\circ = \tan 20^\circ - \left(\frac{ad+bc}{bd-ac}\right)$$

$$\because \tan 20^\circ = \frac{bd-ac}{-(bc+ad)}$$

$$\Rightarrow \left(\frac{a^2+b^2}{bd-ac}\right)\sec 20^\circ = \tan 20^\circ + \frac{1}{\tan 20^\circ} = \frac{\sec^2 20^\circ}{\tan 20^\circ}$$

$$\Rightarrow \left(\frac{a^2+b^2}{bd-ac}\right)\sin 20^\circ = 1 \qquad \text{...(vi)}$$

By (v) + (vi); we get

$$\left(\frac{a^2+b^2+c^2+d^2}{bd-ac}\right)\sin 20^\circ = 2$$

4. In a right angled triangle, acute angles A and B satisfy $\tan A + \tan B + \tan^2 A + \tan^2 B + \tan^3 A + \tan^3 B = 70$. Find the angle A and B in radians.

एक समकोणीय त्रिभुज ABC में कोण A तथा कोण B न्यून कोण है तथा $\tan A + \tan B + \tan^2 A + \tan^2 B + \tan^3 A + \tan^3 B = 70$ को संतुष्ट करते है तो A व B के रेडियन में मान बताओं।

Ans. $\frac{\pi}{12}$ and $\frac{5\pi}{12}$

Sol. Given, in right angled triangle ΔABC; acute angles A and B satisfy
$\tan A + \tan B + \tan^2 A + \tan^2 B + \tan^3 A + \tan^3 B = 70$
$\Rightarrow \tan A + \tan^2 A + \tan^3 A + \tan B + \tan^2 B + \tan^3 B = 70$

$\because A + B = \frac{\pi}{2}$

$\Rightarrow \tan B = \cot A$
$\Rightarrow (\tan A + \cot A) + (\tan^2 A + \cot^2 A) + (\tan^3 A + \cot^3 A) = 70$
$\Rightarrow (\tan A + \cot A) + (\tan A + \cot A)^2 - 2 + (\tan A + \cot A)^3 - 3(\tan A + \cot A) = 70$
$\Rightarrow (\tan A + \cot A)^3 + (\tan A + \cot A)^2 - 2(\tan A + \cot A) = 72$
Let $\tan A + \cot A = x$
$\Rightarrow x^3 + x^2 - 2x - 72 = 0$
$\Rightarrow (x-4)(x^2 + 5x + 18) = 0$
$\Rightarrow$ Only possible root is 4.

$$\Rightarrow \tan A + \frac{1}{\tan A} = 4$$
$$\Rightarrow \tan^2 A - 4\tan A + 1 = 0$$
$$\Rightarrow \tan A = \frac{4 \pm \sqrt{16-4}}{2} = \frac{4 \pm 2\sqrt{3}}{2}$$
$$\Rightarrow \tan A = 2 + \sqrt{3} \text{ or } \Rightarrow \tan A = 2 - \sqrt{3}$$
$\Rightarrow A = 15^\circ$ or 75°
$$\Rightarrow A = \frac{\pi}{12} \text{ or } \frac{5\pi}{12}$$

$\Rightarrow B = \frac{5\pi}{12}$ or $\frac{\pi}{12}$

5. If the product $(\sin 1°)(\sin 3°)(\sin 5°)(\sin 7°) \ldots\ldots\ldots(\sin 89°) = \frac{1}{2^n}$, then find the value of n.

यदि $(\sin 1°)(\sin 3°)(\sin 5°)(\sin 7°) \ldots\ldots\ldots(\sin 89°) = \frac{1}{2^n}$ तो n का मान बताओ।

Ans. $\frac{89}{2}$

Sol. $(\sin 1^o)(\sin 3^o)(\sin 5^o) \ldots\ldots \sin(89^o) = \frac{1}{2^n}$

$$\Rightarrow \sum_{r=1}^{45} \sin\left(\frac{(2r-1)\pi}{180}\right) = \frac{1}{2^n}$$

$$\Rightarrow \frac{(\sin 1^o)(\sin 89^o)(\sin 2^o)(\sin 88^o)\ldots\ldots\ldots\sin 44^o \sin 46^o \cdot \sin 45^o}{\sin 2^o \ldots\ldots\ldots \sin 88^o} = \frac{1}{2^n}$$

$$\Rightarrow \frac{(\sin 1^o \cos 1^o)(\sin 2^o \cos 2^o)\ldots\ldots\ldots(\sin 44^o \cos 44^o)\cdot \sin(45^o)}{(\sin 2^o \sin 4^o \ldots\ldots\ldots \sin 88^o)} = \frac{1}{2^n}$$

$$\Rightarrow \frac{1}{2^{44}} \cdot \frac{1}{\sqrt{2}}\left(\frac{\sin 2^o \sin 4^o \ldots\ldots\ldots \sin 88^o}{\sin 2^o \sin 4^o \ldots\ldots\ldots \sin 88^o}\right) = \frac{1}{2^n}$$

$$\Rightarrow n = 44 + \frac{1}{2} = \frac{89}{2}$$

6. If $\tan\alpha = p/q$ where $\alpha = 6\beta$, α being an acute angle, prove that : $\frac{1}{2}(p\,\text{cosec}\,2\beta - q\sec 2\beta) = \sqrt{p^2+q^2}$.

यदि $\tan\alpha = p/q$ जहां $\alpha = 6\beta$ तथा α एक न्यून कोण हो, तो सिद्ध करो : $\frac{1}{2}(p\,\text{cosec}\,2\beta - q\sec 2\beta) = \sqrt{p^2+q^2}$.

Ans.

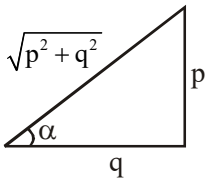
Sol. $\frac{1}{2}\left(\frac{p}{\sin 2\beta} - \frac{q}{\cos 2\beta}\right) = \frac{p\cos 2\beta - q\sin 2\beta}{2\sin 2\beta \cos 2\beta}$

Put $p = \sqrt{p^2+q^2}\ \sin\alpha$ and $q = \sqrt{p^2+q^2}\ \cos\alpha$ using $\tan\alpha = p/q$

$$= \frac{\sqrt{p^2+q^2}\sin\alpha\cos 2\beta - \sqrt{p^2+q^2}\cos\alpha\sin 2\beta}{2\sin 2\beta\cos 2\beta}$$

$$= \sqrt{p^2+q^2}\ \frac{(\sin\alpha\cos 2\beta - \cos\alpha\sin 2\beta)}{\sin 4\beta}$$

$$= \sqrt{p^2+q^2}\ \frac{\sin(\alpha-2\beta)}{\sin 4\beta}$$



$$= \sqrt{p^2+q^2}$$

7. Determine the smallest positive value of x (in degrees) for which $\tan(x+100°) = \tan(x+50°)\tan x\ \tan(x-50°)$.

x का न्यूनतम धनात्मक मान (डिग्री में) ज्ञात करें, जो $\tan(x+100°) = \tan(x+50°)\tan x\ \tan(x-50°)$ को संतुष्ट करता है।

Ans. $x = 30°$

Sol. $\tan(x+100^o) = \tan(x+50^o)\tan x\tan(x-50^o)$

$$\frac{\tan(x+100^o)}{\tan(x-50^o)} = \tan x\tan(x+50^o)$$

$$\Rightarrow \frac{\sin(x+100^o)\cos(x-50^o)}{\cos(x+100^o)\sin(x-50^o)} = \frac{\sin x\sin(x+50^o)}{\cos x\cos(x+50^o)}$$

Apply componendo–dividendo

$$\Rightarrow \frac{\sin(x+100^o)\cos(x-50^o)+\cos(x+100^o)\sin(x-50^o)}{\sin(x+100^o)\cos(x-50^o)-\cos(x+100^o)\sin(x-50^o)} = \frac{\sin x\sin(x+50^o)+\cos x\cos(x+50^o)}{\sin x\sin(x+50^o)-\cos x\cos(x+50^o)}$$

$$\Rightarrow \frac{\sin(x+100^o+x-50^o)}{\sin(x+100^o-x+50^o)} = \frac{\cos(x+50^o-x)}{-\cos(x+50^o+x)}$$

$$\Rightarrow \frac{\sin(2x+50^o)}{\sin(150^o)} = \frac{-\cos(50^o)}{\cos(2x+50^o)}$$

$$\Rightarrow \sin(2x+50^o)\cos(2x+50^o) = -\frac{1}{2}\cos 50^o$$

$$\Rightarrow \sin(4x+100^o) = -\cos(50^o)$$

$$\Rightarrow \sin(4x+100^o) = \sin\left(\frac{3\pi}{2}-50^o\right) = \sin(-40^o)$$

$\Rightarrow 4x + 100^o = n\pi - (-1)^n\, 40^o$

for smallest positive number

substitute n = 1

$4x + 100^o = \pi + 40^o = 220^o$

$\Rightarrow 4x = 120^o$

$\Rightarrow x = 30^o$

8. $\forall\, x \in R$, find the range of the function, $f(x) = \cos x\,(\sin x + \sqrt{\sin^2 x + \sin^2 \alpha}\,)$; $\alpha \in [0, \pi]$

यदि $\forall\, x \in R$ तथा $\alpha \in [0, \pi]$ तो फलन $f(x) = \cos x\,(\sin x + \sqrt{\sin^2 x + \sin^2 \alpha}\,)$ का परिसर बताओं।

Ans. $-\sqrt{1+\sin^2\alpha} \le y \le \sqrt{1+\sin^2\alpha}$

Sol. $y = f(x) = \cos x\left(\sin x + \sqrt{\sin^2 x + \sin^2 \alpha}\right)\ \forall x \in R; \alpha \in [0, \pi]$

$y\sec x - \sin x = \sqrt{\sin^2 x + \sin^2 \alpha}$

$\Rightarrow y^2 \sec^2 x + \sin^2 x - 2\tan xy = \sin^2 x + \sin^2 \alpha$

$\Rightarrow y^2\left(1 + \tan^2 x\right) - 2\tan xy = \sin^2 \alpha$

$\Rightarrow \underbrace{\tan^2 x\left(y^2\right) + \tan x(-2y) + y^2 - \sin^2 \alpha = 0}_{\downarrow\ \tan x \in (-\infty, \infty)}$

$\Rightarrow D \ge 0$ for real values of $\tan x$

$\Rightarrow 4y^2\left(1 - \left(y^2 - \sin^2 \alpha\right)\right) \ge 0$

$\Rightarrow \left(1 + \sin^2 \alpha - y^2\right) \ge 0 \Rightarrow y \in \left[-\sqrt{1+\sin^2\alpha}, \sqrt{1+\sin^2\alpha}\right]$

9. Prove that : $4\sin 27° = \left(5 + \sqrt{5}\right)^{1/2} - \left(3 - \sqrt{5}\right)^{1/2}$.

सिद्ध कीजिए : $4\sin 27° = \left(5 + \sqrt{5}\right)^{1/2} - \left(3 - \sqrt{5}\right)^{1/2}$.

Ans.

Sol. $4\sin 27^\circ = 4\sin(45^\circ - 18^\circ) = 4[\sin(45^\circ)\cos(18^\circ) - \cos(45^\circ)\sin(18^\circ)]$

$$= \frac{4}{\sqrt{2}}\left[\cos(18^\circ) - \sin(18^\circ)\right]$$

$$= \frac{4}{\sqrt{2}}\left[\sqrt{\frac{10+2\sqrt{5}}{16}} - \left(\frac{\sqrt{5}-1}{4}\right)\right]$$

$$= \frac{1}{\sqrt{2}}\left[\sqrt{10+2\sqrt{5}} - \left(\sqrt{5}-1\right)\right]$$

$$= \left[\sqrt{5+\sqrt{5}} - \left(\frac{\sqrt{\left(\sqrt{5}-1\right)^2}}{\sqrt{2}}\right)\right]$$

$$= \left(5+\sqrt{5}\right)^{\frac{1}{2}}\left(3-\sqrt{5}\right)^{\frac{1}{2}}$$

10. Let $x_1 = \prod_{r=1}^{5}\cos\frac{r\pi}{11}$ and $x_2 = \sum_{r=1}^{5}\cos\frac{r\pi}{11}$, then show that $x_1 \cdot x_2 = \frac{1}{64}\left(\operatorname{cosec}\frac{\pi}{22} - 1\right)$, where Π denotes the continued product.

यदि $x_1 = \prod_{r=1}^{5}\cos\frac{r\pi}{11}$ तथा $x_2 = \sum_{r=1}^{5}\cos\frac{r\pi}{11}$, तो सिद्ध करो कि $x_1 \cdot x_2 = \frac{1}{64}\left(\operatorname{cosec}\frac{\pi}{22} - 1\right)$, जहां Π सतत गुणन (continued product) है।

Ans.

Sol. $x_1 = \prod_{r=1}^{5}\cos\left(\frac{r\pi}{11}\right) = \frac{\cos\pi}{11}\cdot\frac{\cos 2\pi}{11}\cdot\frac{\cos 3\pi}{11}\cdot\frac{\cos 4\pi}{11}\cdot\frac{\cos 5\pi}{11}$

$$\Rightarrow x_1 = \cos\frac{\pi}{11}\cdot\cos\frac{2\pi}{11}\cdot\cos\frac{4\pi}{11}\cdot\cos\frac{8\pi}{11}\cdot\cos\frac{5\pi}{11}$$

$$\Rightarrow x_1 = \frac{1}{16}\sin\left(\frac{16\pi}{11}\right)\cdot\frac{1}{\sin\left(\frac{\pi}{11}\right)}\cdot\cos\left(\frac{5\pi}{11}\right)$$

$$\Rightarrow x_1 = \frac{1}{16}\frac{\sin\left(\frac{5\pi}{11}\right)\cdot\cos\left(\frac{5\pi}{11}\right)}{\sin\left(\frac{\pi}{11}\right)} = \frac{1}{32}\frac{\sin\left(\frac{10\pi}{11}\right)}{\sin\left(\frac{\pi}{11}\right)} = \frac{1}{32}$$

$$x_2 = \sum_{r=1}^{5} \cos\left(\frac{r\pi}{11}\right) = \cos\left(\frac{\pi}{11}\right) + \cos\left(\frac{2\pi}{11}\right) + ... + \cos\left(\frac{5\pi}{11}\right)$$

$$= \cos\left(\frac{\frac{\pi}{11} + \frac{5\pi}{11}}{2}\right) \frac{\sin\left(\frac{5\pi}{22}\right)}{\sin\left(\frac{\pi}{22}\right)}$$

$$\Rightarrow x_2 = \frac{\cos\left(\frac{6\pi}{22}\right) \cdot \sin\left(\frac{5\pi}{22}\right)}{\sin\left(\frac{\pi}{22}\right)}$$

$$\Rightarrow x_2 = \frac{\sin\left(\frac{5\pi}{22} + \frac{6\pi}{22}\right) + \sin\left(\frac{5\pi}{22} - \frac{6\pi}{22}\right)}{2\sin\left(\frac{\pi}{22}\right)}$$

$$\Rightarrow x_2 = \frac{1 - \sin\left(\frac{\pi}{22}\right)}{2\sin\left(\frac{\pi}{22}\right)} = \frac{1}{2}\left[\operatorname{cosec}\left(\frac{\pi}{22}\right) - 1\right]$$

$$\Rightarrow x_1 \cdot x_2 = \frac{1}{64}\left(\operatorname{cosec}\left(\frac{\pi}{22}\right) - 1\right)$$

11. If $(1 + \sin t)(1 + \cos t) = \frac{5}{4}$. Find the value of $(1 - \sin t)(1 - \cos t)$.

यदि $(1 + \sin t)(1 + \cos t) = \frac{5}{4}$ तो $(1 - \sin t)(1 - \cos t)$ का मान बताओं।

Ans. $\frac{13}{4} - \sqrt{10}$

Sol. Given

$$(1 + \sin t)(1 + \cos t) = \frac{5}{4}$$

$$\Rightarrow 1 + \sin t + \cos t + \sin t \cos t = \frac{5}{4}$$

$$\Rightarrow 2+2\sin t\cos t+2\left(\sin t+\cos t\right)=\frac{5}{2}$$

$$\Rightarrow 1+2\sin t\cos t+2\left(\sin t+\cos t\right)=\frac{3}{2}$$

$$\Rightarrow \sin^2 t+\cos^2 t+2\sin t\cos t+2\left(\sin t+\cos t\right)=\frac{3}{2}$$

$$\Rightarrow \left(\sin t+\cos t\right)^2+2\left(\sin t+\cos t\right)-\frac{3}{2}=0$$

$$\Rightarrow \sin t+\cos t=\frac{-2\pm\sqrt{4+6}}{2}$$

$$\because \sin t+\cos t\in\left[-\sqrt{2},\sqrt{2}\right]$$

$$\Rightarrow \sin t+\cos t=\frac{\sqrt{10}}{2}-1$$

$\therefore (1-\sin t)(1-\cos t)=1-(\sin t+\cos t)+\sin t\cos t$

$$=\frac{5}{4}-2\left(\sin t+\cos t\right)=\frac{5}{4}-\sqrt{10}+2$$

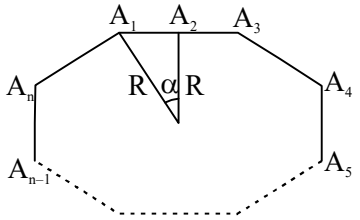
$$=\frac{13}{4}-\sqrt{10}$$

12. Let $A_1, A_2, A_3 \ldots\ldots\ldots A_n$ are the vertices of a regular n sided polygon inscribed in a circle of radius R. If $(A_1A_2)^2+(A_1A_3)^2+\ldots\ldots\ldots+(A_1A_n)^2=14R^2$, find the number of sides in the polygon.

यदि $A_1, A_2, A_3 \ldots\ldots\ldots A_n$, n भुजाओं वाले समबाहुभुज (Regular polygon) के शीर्ष है, जिसके परिवृत की त्रिज्या R है। यदि $(A_1A_2)^2+(A_1A_3)^2+\ldots\ldots\ldots+(A_1A_n)^2=14R^2$ तो बहुभुज में भुजाओं की संख्या बताओं।

Ans. n = 7

Sol. $\alpha=\frac{\pi}{n}$



$A_1A_2=2R\sin\alpha$

$A_1A_3=2R\sin 2\alpha$

$A_1A_4=2R\sin 3\alpha$

$\vdots$

$A_1A_n=2R\sin(n-1)\alpha$

$$\Rightarrow \left(A_1A_2\right)^2+\left(A_1A_3\right)^2+\ldots\ldots\ldots+\left(A_1A_n\right)^2$$

$$=4R^2\left[\sin^2\alpha+\sin^2 2\alpha+\ldots\ldots\ldots+\sin^2\left(n-1\right)\alpha\right]$$

$$= 2R^2\left[1-\cos 2\alpha+1-\cos 4\alpha+1-\cos 6\alpha+.....+1-\cos 2(n-1)\alpha\right]$$

$$= 2R^2\left[(n-1)-\left[\cos(2\alpha)+\cos(4\alpha)+\cos 6\alpha+.....+\cos 2(n-1)\alpha\right]\right]$$

$$= 2R^2\left[(n-1)-\frac{\sin\left(\frac{(n-1)2\alpha}{2}\right)}{\sin\left(\frac{2\alpha}{2}\right)}\cdot\cos\left(\frac{2n\alpha}{2}\right)\right]$$

$$= 2R^2\left[(n-1)-\frac{\sin((n-1)\alpha)}{\sin\alpha}\cdot\cos(n\alpha)\right]$$

$$= 2R^2\,[(n-1)+1] = 14R^2 \quad \left\{\because \alpha = \frac{\pi}{n}\right\} = 2nR^2$$

$$\Rightarrow n = 7$$

13. Let $\theta \in R$ and $\sum_{k=2}^{\infty} \sin\left(2^k\theta\right) = a$. Find the value $\sum_{k=0}^{\infty} \left[\cot^3\left(2^k\theta\right)-\cot\left(2^k\theta\right)\right]\sin^4\left(2^k\theta\right)$ in terms of 'a'.

यदि $\sum_{k=2}^{\infty} \sin\left(2^k\theta\right) = a$ जहां $\theta \in R$ तो व्यंजक $\sum_{k=0}^{\infty} \left[\cot^3\left(2^k\theta\right)-\cot\left(2^k\theta\right)\right]\sin^4\left(2^k\theta\right)$ का मान 'a' के पदों में बताओं।

Ans. $\frac{a}{4}$

Sol. Given

$$\theta \in R \ \& \ \sum_{k=2} \sin\left(2^k\theta\right) = a$$

$$\sum_{k=0}^{\infty}\left[\cot^3\left(2^k\theta\right)-\cot\left(2^k\theta\right)\right]\sin^4\left(2^k\theta\right)$$

$$= \sum_{k=0}^{\infty}\cot\left(2^k\theta\right)\left(\cot^2\left(2^k\theta\right)-1\right)\cdot\sin^4\left(2^k\theta\right)$$

$$= \sum_{k=0}^{\infty}\cot\left(2^k\theta\right)\cdot\frac{\left(\cos^2\left(2^k\theta\right)-\sin^2\left(2^k\theta\right)\right)}{\sin^2\left(2^k\theta\right)}\cdot\sin^4\left(2^k\theta\right)$$

$$= \sum_{k=0}^{\infty} \cos\left(2^k\theta\right)\cdot\sin\left(2^k\theta\right)\cdot\cos\left(2^{k+1}\theta\right)$$

$$= \frac{1}{4}\sum_{k=0}^{\infty}\sin\left(2^{k+2}\theta\right) = \frac{1}{4}\sum_{k=2}^{\infty}\sin\left(2^k\theta\right)$$

$$= \frac{a}{4}$$

14. If x and y are real number such that $x^2 + 2xy - y^2 = 6$, find the minimum value of $(x^2 + y^2)^2$.

यदि वास्तविक संख्याऐं x तथा y सम्बन्ध $x^2 + 2xy - y^2 = 6$ को संतुष्ट करती है तो $(x^2 + y^2)^2$ का न्यूनतम मान बताओ।

Ans. 18

Sol. $x^2 + 2xy - y^2 = 6$

$(x^2 - y^2)^2 = (6 - 2xy)^2$

$\Rightarrow x^4 + y^4 - 2x^2y^2 + 4x^2y^2 = 36 + 4x^2y^2 + 4x^2y^2 - 24xy$

$\Rightarrow (x^2 + y^2)^2 = 8x^2y^2 - 24xy + 36$

$\Rightarrow (x^2 + y^2)^2 = 4\,(2x^2y^2 - 6xy + 9)$

$$\underbrace{2(xy)^2 - 6(xy) + 9 = 0}_{\text{quad in xy}}$$

Minimum value is $\left(\frac{-D}{4a}\right) = -\left(\frac{36-72}{8}\right) = \frac{36}{8}$

$\Rightarrow$ Minimum value of $(x^2 + y^2)^2$ is $4 \times \frac{36}{8} = 18$

15. Find the exact value of $\tan^2\frac{\pi}{16} + \tan^2\frac{3\pi}{16} + \tan^2\frac{5\pi}{16} + \tan^2\frac{7\pi}{16}$

$\tan^2\frac{\pi}{16} + \tan^2\frac{3\pi}{16} + \tan^2\frac{5\pi}{16} + \tan^2\frac{7\pi}{16}$ का आंकिक मान बताओं।

Ans. 28

Sol. $\tan^2\frac{\pi}{16} + \tan^2\frac{3\pi}{16} + \tan^2\frac{5\pi}{16} + \tan^2\frac{7\pi}{16}$

$= \tan^2\frac{\pi}{16} + \tan^2\frac{7\pi}{16} + \tan^2\frac{3\pi}{16} + \tan^2\frac{5\pi}{16}$

$$= \tan^2\frac{\pi}{16} + \tan^2\left(\frac{\pi}{2} - \frac{\pi}{16}\right) + \tan^2\left(\frac{3\pi}{16}\right) + \tan^2\left(\frac{\pi}{2} - \frac{3\pi}{16}\right)$$

$$= \tan^2\frac{\pi}{16} + \cot^2\frac{\pi}{16} + \tan^2\frac{3\pi}{16} + \cot^2\frac{3\pi}{16}$$

$$= \left(\tan\frac{\pi}{16} + \cot\frac{\pi}{16}\right)^2 + \left(\tan\frac{3\pi}{16} + \cot\frac{3\pi}{16}\right)^2 - 4$$

$$= \left(2\,\text{cosec}\left(\frac{2\pi}{16}\right)\right)^2 + \left(2\,\text{cosec}\left(\frac{3\pi}{18}\right)\right)^2 - 4$$

$$= 4\left[\text{cosec}^2\frac{\pi}{8} + \text{cosec}^2\frac{3\pi}{8}\right] - 4$$

$$= 4\left[\text{cosec}^2\frac{\pi}{8} + \text{cosec}^2\left(\frac{\pi}{2} - \frac{3\pi}{8}\right)\right] - 4$$

$$= 4\left[\text{cosec}^2\frac{\pi}{8} + \text{sec}^2\frac{\pi}{8}\right] - 4$$

$$= 4\left[\frac{1}{\sin^2\frac{\pi}{8}\cdot\cos^2\frac{\pi}{8}}\right] - 4$$

$$= 16\left[\frac{1}{\sin^2\frac{\pi}{4}}\right] - 4$$

$= 32 - 4 = 28$

16. If 'θ' is eliminated from the equations $\cos\theta - \sin\theta = b$ and $\cos 3\theta + \sin 3\theta = a$, find the eliminant.

यदि समीकरण $\cos\theta - \sin\theta = b$ तथा $\cos 3\theta + \sin 3\theta = a$ में से 'θ' विलुप्त किया जाये, तो a तथा b में प्रतिबन्ध ज्ञात किजिए।

Ans. $a = 3b - 2b^3$

Sol. $\cos\theta - \sin\theta = b$

$\cos 3\theta + \sin 3\theta = a$

$\Rightarrow a = \cos 3\theta + \sin 3\theta$

$= 4\cos^3\theta - 3\cos\theta + 3\sin\theta - 4\sin^3\theta$

$= 4\,[\cos^3\theta - \sin^3\theta] + 3\,(\sin\theta - \cos\theta)$

$= 4\,[(\cos\theta - \sin\theta)^3 + 3\sin\theta\cos\theta\,(\cos\theta - \sin\theta)] + (\sin\theta - \cos\theta)$

$= 4\,[b^3 + 3\sin\theta\cos\theta\,(b)] + - 3b$

$\Rightarrow a = 4b^3 - 3b + 12b \sin\theta \cos\theta$

$\because \cos\theta - \sin\theta = b$

$\Rightarrow \cos^2\theta + \sin^2\theta - \sin 2\theta = b^2$

$$\Rightarrow a = 4b^3 - 3b + 12b\left(\frac{1-b^2}{2}\right)$$

$$\Rightarrow a = 4b^3 - 3b + 6b - 6b^3$$

$$\Rightarrow a = 3b - 2b^3$$

17. Show that elliminating x and y from the equations, $\sin x + \sin y = a$; $\cos x + \cos y = b$ and $\tan x + \tan y = c$ gives

$$\frac{8ab}{\left(a^2+b^2\right)^2 - 4a^2} = c.$$

सिद्ध करिये कि समीकरण $\sin x + \sin y = a$; $\cos x + \cos y = b$ तथा $\tan x + \tan y = c$ से x व y विलुप्त करने पर $\frac{8ab}{\left(a^2+b^2\right)^2 - 4a^2}$ = c प्राप्त होगा।

Ans.

Sol. Given

$\sin x + \sin y = a$

$\cos x + \cos y = b$

$\tan x + \tan y = c$

$\Rightarrow \sin^2 x + \sin^2 y + 2 \sin x \sin y = a^2$ ...(i)

$\cos^2 x + \cos^2 y + 2 \cos x \cos y = b^2$ ...(ii)

By (i) + (ii); we get

$1 + 1 + 2 (\sin x \sin y + \cos x \cos y) = a^2 + b^2$

$$\Rightarrow a^2 + b^2 = 2 + 2\cos(x - y) = 4\cos^2\left(\frac{x-y}{2}\right)$$

Now $\tan x + \tan y = c$

$$\Rightarrow \frac{\sin x \cos y + \cos x \sin y}{\cos x \cos y} = c$$

$$\because a = \sin x + \sin y = 2\sin\left(\frac{x+y}{2}\right)\cos\left(\frac{x-y}{2}\right)$$

$$\Rightarrow \left(a^2 + b^2\right)^2 - 4a^2 = 16\cos^4\left(\frac{x-y}{2}\right) - 16\sin^2\left(\frac{x+y}{2}\right)\cos^2\left(\frac{x-y}{2}\right)$$

$$= 16\cos^2\left(\frac{x-y}{2}\right)\left(\cos^2\left(\frac{x-y}{2}\right) - \sin^2\left(\frac{x+y}{2}\right)\right)$$

$$\left(a^2+b^2\right)^2-4a^2=16\cos^2\left(\frac{x-y}{2}\right)\cos x\cos y \qquad ...(i)$$

Now

$2ab = 2 (\sin x + \sin y) (\cos x + \cos y)$

$= \sin 2x + \sin 2y + 2 \sin (x + y)$

$= 2 \sin (x + y) \cos (x - y) + 2 \sin (x + y)$

$$= 4\sin\left(x+y\right)\cos^2\left(\frac{x-y}{2}\right)$$

$$\Rightarrow 8ab=16\sin\left(x+y\right)\cos^2\left(\frac{x-y}{2}\right) \qquad ...(ii)$$

By (i) & (ii)

$$\frac{8ab}{\left(a^2+b^2\right)^2-4a^2}=\frac{16\sin\left(x+y\right)\cos^2\left(\frac{x-y}{2}\right)}{16\cos^2\left(\frac{x-y}{2}\right)\cos x\cos y}$$

$= \tan x + \tan y = c$

18. Given that $3 \sin x + 4 \cos x = 5$ where $x \in (0, \pi/2)$. Find the value of $2 \sin x + \cos x + 4 \tan x$.

दिया है कि $3 \sin x + 4 \cos x = 5$ जहां $x \in (0, \pi/2)$ तो $2 \sin x + \cos x + 4 \tan x$ का मान बताओं।

Ans. 5

Sol. Given

$$3 \sin x + 4 \cos x = 5;\ x \in \left(0, \frac{\pi}{2}\right)$$

$$\Rightarrow 3\sin x = 5 - 4\cos x$$

Square both sides

$\Rightarrow 9 \sin^2 x = (5 - 4 \cos x)^2 = 25 + 16 \cos^2 x - 40 \cos x$

$\Rightarrow 9 (1 - \cos^2 x) = 25 + 16 \cos^2 x - 40 \cos x$

$\Rightarrow 25 \cos^2 x - 40 \cos x + 16 = 0$

$$\Rightarrow \cos x = \frac{40 \pm \sqrt{1600 - 1600}}{50}$$

$$\Rightarrow \cos x = \frac{4}{5}$$

$$\Rightarrow \sin x = \frac{3}{5}$$

$$\Rightarrow \tan x = \frac{3}{4}$$

$$\Rightarrow 2\sin x + \cos x + 4\tan x = \frac{6}{5} + \frac{4}{5} + 3 = 5$$

19. Prove that the triangle ABC is equilateral iff, $\cot A + \cot B + \cot C = \sqrt{3}$.

यदि त्रिभुज ABC एक समबाहु त्रिभुज हो तो यदि और केवल यदि कि $\cot A + \cot B + \cot C = \sqrt{3}$.

Ans.

Sol. If ΔABC is equilateral then $A = B = C = \frac{\pi}{3}$

$$\therefore \cot\frac{\pi}{3} + \cot\frac{\pi}{3} + \cot\frac{\pi}{3} = \sqrt{3}$$

Now if $\cot A + \cot B + \cot C = \sqrt{3}$
sq. both sides we get
$\Sigma \cot^2 A + 2\Sigma \cot A \cot B = 3(1)$
we know that $\Sigma \cot A \cot B = 1$
$\therefore \Sigma \cot^2 A + 2\Sigma \cot A \cot B = 3\ (\Sigma \cot A \cot B)$
$\Rightarrow \Sigma \cot^2 A - \Sigma \cot A \cot B = 0$
$\Rightarrow (\cot A - \cot B)^2 + (\cot B - \cot C)^2 + (\cot C - \cot A)^2 = 0$
$\Rightarrow \cot A = \cot B = \cot C \Rightarrow A = B = C$
$\Rightarrow \Delta ABC$ is equilateral.

20. If $A+B+C = \pi$; prove that $\tan^2\frac{A}{2} + \tan^2\frac{B}{2} + \tan^2\frac{C}{2} \geq 1$.

यदि $A+B+C = \pi$; तो सिद्ध करो कि $\tan^2\frac{A}{2} + \tan^2\frac{B}{2} + \tan^2\frac{C}{2} \geq 1$.

Ans.

Sol. Given $A + B + C = \pi$

$$\Rightarrow \frac{A}{2} + \frac{B}{2} + \frac{C}{2} = \frac{\pi}{2}$$

$$\Rightarrow \tan\frac{A}{2}\tan\frac{B}{2} + \tan\frac{B}{2}\tan\frac{C}{2} + \tan\frac{C}{2}\tan\frac{A}{2} = I$$

$$\because \left(\tan\frac{A}{2} - -\tan\frac{B}{2}\right)^2 \geq 0 \qquad ...(i)$$

$$\left(\tan\frac{B}{2} - \tan\frac{C}{2}\right)^2 \geq 0 \qquad ...(ii)$$

$$\left(\tan\frac{C}{2}-\tan\frac{A}{2}\right)^2 \geq 0 \qquad \text{...(iii)}$$

By (i) + (ii) + (iii); we get

$$2\left(\tan^2\frac{A}{2}+\tan^2\frac{B}{2}+\tan^2\frac{C}{2}\right) \geq 2\sum\tan\frac{A}{2}\cdot\tan\frac{B}{2}$$

$$\Rightarrow \tan^2\frac{A}{2}+\tan^2\frac{B}{2}+\tan^2\frac{C}{2} \geq 1$$

21. If $\tan\left(\frac{\pi}{4}+\frac{y}{2}\right)=\tan^3\left(\frac{\pi}{4}+\frac{x}{2}\right)$, prove that $\frac{\sin y}{\sin x}=\frac{3+\sin^2 x}{1+3\sin^2 x}$.

यदि $\tan\left(\frac{\pi}{4}+\frac{y}{2}\right)=\tan^3\left(\frac{\pi}{4}+\frac{x}{2}\right)$, तो सिद्ध करो कि $\frac{\sin y}{\sin x}=\frac{3+\sin^2 x}{1+3\sin^2 x}$.

Ans.

Sol. $$\tan\left(\frac{\pi}{4}+\frac{y}{2}\right)=\sqrt{\frac{1-\cos\left(\frac{\pi}{2}+y\right)}{1+\cos\left(\frac{\pi}{2}+y\right)}}==\sqrt{\frac{1+\sin y}{1-\sin y}}$$

$$\Rightarrow \frac{1+\sin y}{1-\sin y}=\left(\frac{1+\sin x}{1-\sin x}\right)^3 \Rightarrow \frac{\sin y+1}{\sin y-1}=\left(\frac{\sin x+1}{\sin x-1}\right)^3$$

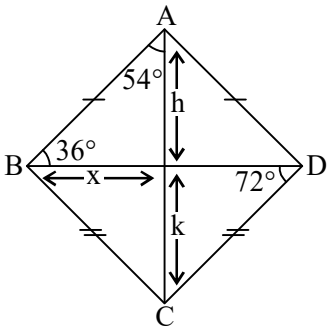
Now apply C and D

$$\Rightarrow \frac{\sin y}{1}=\frac{2\sin^3 x+6\sin x}{2+6\sin^2 x}=\frac{\sin x\left(3+\sin^2 x\right)}{1+3\sin^2 x}$$

22. In a kite ABCD, AB = AD and CB = CD. If $\angle A = 108°$ and $\angle C = 36°$ then the ratio of the area of ΔABD to the area of ΔCBD can be written in the form $\frac{a-b\tan^2 36°}{c}$ where a, b and c are relatively prime positive integers.determine the ordered triple (a, b, c). Also find the numerical value of the ratio. एक पतंग (kite) ABCD में AB = AD तथा CB = CD है। यदि $\angle A = 108°$ तथा $\angle C = 36°$ तो ΔABD व ΔCBD के क्षेत्रफलों का अनुपात $\frac{a-b\tan^2 36°}{c}$ के रूप में लिखा जा सकता है जहाँ a, b तथा c सापेक्षिक अभाज्य (relatively prime) संख्याऐं हो, तो a, b, c का मान बताओं तथा उपरोक्त अनुपात का आंकिक मान भी बताओं।

Ans. $(1,1,2);\ \sqrt{5}-2$

Sol. Since ΔABD and ΔCBD have a common base.
$\Rightarrow$ ratio of their areas equals the ratio of their heights.



$\because \tan 36^\circ = \dfrac{h}{x}$ & $\tan 72^\circ = \dfrac{k}{x}$

$$\Rightarrow \frac{h}{k} = \frac{x\tan 36^\circ}{x\tan 72^\circ} = \frac{\tan 36^\circ}{2\tan 36^\circ}\left(1-\tan^2 36^\circ\right)$$

$$\Rightarrow \frac{h}{k} = \frac{1-\tan^2 36^\circ}{2}$$

$\Rightarrow a = 1, b = 1, c = 2$

$$\text{Now } \frac{1-\tan^2 36^\circ}{2} = \frac{\cos^2 36^\circ - \sin^2 36^\circ}{2\cos^2 36^\circ}$$

$$= \frac{\cos 72^\circ}{2\cos^2 36^\circ}$$

$$= \frac{\cos 72^\circ}{1+\cos 72^\circ} = \frac{\sin 18^\circ}{1+\sin 18^\circ} = \frac{\sqrt{5}-1}{4+\sqrt{5}-1}$$

$$= \frac{\sqrt{5}-1}{\sqrt{5}+3} = \frac{\left(\sqrt{5}-1\right)\left(\sqrt{5}-3\right)}{-4}$$

$$= \sqrt{5}-2$$

23. If $4\sin x \cdot \cos y + 2\sin x + 2\cos y + 1 = 0$ where $x, y \in [0, 2\pi]$ find the largest possible value of the sum $(x + y)$.

यदि $4\sin x \cdot \cos y + 2\sin x + 2\cos y + 1 = 0$ जहाँ $x, y \in [0, 2\pi]$ तो $(x + y)$ का अधिकतम सम्भव मान ज्ञात कीजिए।

Ans. $\dfrac{19\pi}{6}$

Sol. $4\sin x\cos y + 2\sin x + 2\cos y + 1 = 0;\ x, y \in [0, 2\pi]$
$2[\sin x + \cos y + 2\sin x\cos y] + 1 = 0$
$\Rightarrow (2\sin x + 1)(2\cos y + 1) = 0$

$$\Rightarrow \sin x = \frac{-1}{2} \text{ or } \cos y = \frac{-1}{2}$$

$$\Rightarrow x = \frac{7\pi}{6}, \frac{11\pi}{6} \text{ or } y = \frac{2\pi}{3}, \frac{4\pi}{3}$$

$\therefore$ Largest possible value of $x + y$ is $\frac{11\pi}{6} + \frac{4\pi}{3} = \frac{19\pi}{6}$

24. Let $x > 1, y > 1$ and $(\ell n\, x)^2 + (\ell n\, y)^2 = \ell n\, x^2 + \ell n\, y^2$, then find the maximum value of $x^{\ell n\, y}$.

माना $x > 1, y > 1$ तथा $(\ell n\, x)^2 + (\ell n\, y)^2 = \ell n\, x^2 + \ell n\, y^2$ तो $x^{\ell n\, y}$ का अधिकतम मान ज्ञात कीजिए।

Ans. e^4

Sol. $(\ell nx)^2 + (\ell ny)^2 = \ell nx^2 + \ell ny^2$

$\Rightarrow (\ell nx)^2 + (\ell n(y))^2 = 2(\ell nx + \ell ny) \qquad \{\because x > 1, y > 1\}$

$\Rightarrow (\ell nx - 1)^2 + (\ell ny - 1)^2 = 2$

$\Rightarrow \left(\frac{\ell nx - 1}{\sqrt{2}}\right)^2 + \left(\frac{\ell ny - 1}{\sqrt{2}}\right)^2 = 1$

Let $\frac{\ell nx - 1}{\sqrt{2}} = \sin\theta$ & $\frac{\ell ny - 1}{\sqrt{2}} = \cos\theta$

$\Rightarrow \ell nx = 1 + \sqrt{2}\sin\theta$ & $\Rightarrow \ell ny = 1 + \sqrt{2}\cos\theta$

$\Rightarrow x^{\ell ny} = e^{(1+\sqrt{2}\sin\theta)(1+\sqrt{2}\cos\theta)}$

$= e^{1+2\sin\theta\cos\theta+\sqrt{2}\cos\theta+\sqrt{2}\sin\theta}$

$\because \cos\theta + \sin\theta \in \left[-\sqrt{2}, \sqrt{2}\right]$

$\Rightarrow$ Max value of $x^{\ell ny} = e^{1+1+2} = e^4$

25. Prove that $1 + \cot\theta \le \cot\frac{\theta}{2}$ for $0 < \theta < \pi$. Find θ when equality sign holds.

सिद्ध कीजिए $1 + \cot\theta \le \cot\frac{\theta}{2}$ for $0 < \theta < \pi$. θ का मान ज्ञात करों जब बराबर का चिन्ह है।

Ans. $\frac{\pi}{2}$

Sol. $1 + \cot\theta - \cot\frac{\theta}{2}$

$$= 1 + \frac{\cot^2\frac{\theta}{2} - 1}{2\cot\frac{\theta}{2}} - \cot\frac{\theta}{2}$$

$$= \frac{2\cot\frac{\theta}{2} - 2\cot^2\frac{\theta}{2} + \cot^2\frac{\theta}{2} - 1}{2\cot\frac{\theta}{2}}$$

$$= -\frac{\left(\cot\frac{\theta}{2} - 1\right)^2}{2\cot\frac{\theta}{2}} \le 0. \text{ for } 0 < \theta < \pi.$$

$$\Rightarrow 1 + \cot\theta \le \cot\frac{\theta}{2}$$

Equality holds if

$$\cot\frac{\theta}{2} = 1$$

$$\Rightarrow \frac{\theta}{2} = \frac{\pi}{4}$$

$$\Rightarrow \theta = \frac{\pi}{2}$$

26. For all θ in $\left[0, \frac{\pi}{2}\right]$, show that $\cos(\sin\theta) > \sin(\cos\theta)$.

सिद्ध कीजिए $\cos(\sin\theta) > \sin(\cos\theta)$, $\forall\ \theta \in \left[0, \frac{\pi}{2}\right]$

Ans.

Sol. $\because \cos\theta + \sin\theta = \sqrt{2}\left[\frac{1}{\sqrt{2}}\cos\theta + \frac{1}{\sqrt{2}}\sin\theta\right]$

$$= \sqrt{2}\left[\sin\left(\frac{\pi}{4} + \theta\right)\right]$$

$$\Rightarrow \cos\theta + \sin\theta \le \sqrt{2} < \frac{\pi}{2}$$

$\Rightarrow \cos\theta < \frac{\pi}{2} - \sin\theta$

As $\theta \in \left[0, \frac{\pi}{2}\right]$ in which sinθ increases.

$\Rightarrow \sin(\cos\theta) < \sin\left(\frac{\pi}{2} - \sin\theta\right)$

$\Rightarrow \cos(\sin\theta) > \sin(\cos\theta) \forall \theta \in \left[0, \frac{\pi}{2}\right]$

27. If $x + y + z = xyz$

Prove that $\frac{x}{1-x^2} + \frac{y}{1-y^2} + \frac{z}{1-z^2} = \frac{4xyz}{(1-x^2)(1-y^2)(1-z^2)}$.

यदि $x + y + z = xyz$

सिद्ध कीजिए $\frac{x}{1-x^2} + \frac{y}{1-y^2} + \frac{z}{1-z^2} = \frac{4xyz}{(1-x^2)(1-y^2)(1-z^2)}$.

Ans.

Sol. $x + y + z = xyz$

$$\frac{x}{1-x^2} + \frac{y}{1-y^2} + \frac{z}{1-z^2} = \frac{x(1-y^2)(1-z^2) + y(1-x^2)(1-z^2) + z(1-x^2)(1-y^2)}{(1-x^2)(1-y^2)(1-z^2)}$$

$$= \frac{x\left[1-y^2-z^2+y^2z^2\right] + y\left[1-x^2-z^2+x^2z^2\right] + z\left[1-x^2-y^2+x^2y^2\right]}{(1-x^2)(1-y^2)(1-z^2)}$$

$$= \frac{x+y+z-xy^2-xz^2+xy^2z^2-x^2y-z^2y+x^2yz^2+x^2y^2z-x^2z-y^2z}{(1-x^2)(1-y^2)(1-z^2)}$$

$$= \frac{x+y+z-xy^2+xz^2y^2+x^2y^2z-y^2z-x^2y+x^2yz^2-x^2z-xz^2-z^2y}{(1-x^2)(1-y^2)(1-z^2)}$$

$$= \frac{x+y+z-y^2(x+z)+xyz(yz+xy)-x^2(y+z)+z^2(x^2y-x-y)}{(1-x^2)(1-y^2)(1-z^2)}$$

$$= \frac{xyz - y^2(xyz - y) + xyz\left(xy^2z - y\right) - x^2(xyz - x) + z^2\left(x^2y + z - xyz\right)}{\left(1-x^2\right)\left(1-y^2\right)\left(1-z^2\right)}$$

$$= \frac{xyz\left[1 - y^2 + xy^2z - x^2 - z^2\right] + \left[y^3 - xy^3z + x^3 + x^2yz^2 + z^3\right]}{\left(1-x^2\right)\left(1-y^2\right)\left(1-z^2\right)}$$

$$= \frac{xyz\left[1 + xy^2z - \left(x^2 + y^2 + z^2\right)\right] + \left[y^3 + x^3 + z^3 - xy^3z + x^2yz^2\right]}{\left(1-x^2\right)\left(1-y^2\right)\left(1-z^2\right)}$$

$$= \frac{xyz\left[1 + xy^2z - \left[x^2y^2z^2 - 2xy - 2xz - 2yz\right]\right] + \left[x^3y^3z^3 - 3x^2y^2z - 3x^2yz^2 - 3xy^2z^2 + 3xyz\right] - xy^3z + x^2yz^2}{\left(1-x^2\right)\left(1-y^2\right)\left(1-z^2\right)}$$

$$= \frac{xyz + x^2y^3z^2 - x^3y^3z^3 + 2x^2y^2z + 2x^2yz^2 + 2y^2z^2x + x^3y^3z^3 - 3x^2y^2z - 3x^2yz^2 - 3xy^2z^2 + 3xyz - xy^3z + x^2yz^2}{\left(1-x^2\right)\left(1-y^2\right)\left(1-z^2\right)}$$

$$= \frac{4xyz + x^2y^3z^2 - x^2y^2z - x^2yz^2 - xy^2z^2 - xy^3z + x^2yz^2}{\left(1-x^2\right)\left(1-y^2\right)\left(1-z^2\right)}$$

$$= \frac{4xyz + x^2y^3z^2 - x^2y^2z - xy^2z^2 - xy^3z}{\left(1-x^2\right)\left(1-y^2\right)\left(1-z^2\right)}$$

$$= \frac{4xyz + xyz\left[xy^2z - xy - yz - y^2\right]}{\left(1-x^2\right)\left(1-y^2\right)\left(1-z^2\right)}$$

$$= \frac{4xyz + xyz\left[y(x + y + z) - xy - yz - y^2\right]}{\left(1-x^2\right)\left(1-y^2\right)\left(1-z^2\right)}$$

$$= \frac{4xyz}{\left(1-x^2\right)\left(1-y^2\right)\left(1-z^2\right)}$$

Method–IInd

$x + y + z = xyz$

Let $x = \tan A$, $y = \tan B$, $z = \tan C$

$\Rightarrow A + B + C = \pi$

$\Rightarrow (2A + 2B + 2C) = 2\pi$

$\Rightarrow \tan(2A + 2B) = \tan(2\pi - 2C)$

$$\frac{\tan 2A + \tan 2B}{1 - \tan 2A \tan 2B} = -\tan 2C$$

$$\Rightarrow \tan 2A + \tan 2B = -\tan 2C + \tan 2A \tan 2B \tan 2C$$

$$\Rightarrow \frac{2\tan A}{1-\tan^2 A} + \frac{2\tan B}{1-\tan^2 B} + \frac{2\tan C}{1-\tan^2 C} = \frac{8\tan A \tan B \tan C}{\left(1-\tan^2 A\right)\left(1-\tan^2 B\right)\left(1-\tan^2 C\right)}$$

$$\Rightarrow \frac{x}{1-x^2} + \frac{y}{1-y^2} + \frac{z}{1-z^2} = \frac{4xyz}{\left(1-x^2\right)\left(1-y^2\right)\left(1-z^2\right)}$$

COMPREHENSION TYPE QUESTIONS

Paragraph for Question No. 28 to 30

If $\cos\alpha + \cos\beta = a$ and $\sin\alpha + \sin\beta = b$ and θ is arithmatic mean between α and β, then

$$\sin 2\theta + \cos 2\theta = 1 + \frac{nb(a-b)}{a^2+b^2}$$

where n is some integer then answer the following questions :

यदि $\cos\alpha + \cos\beta = a$ तथा $\sin\alpha + \sin\beta = b$ एवं θ, α तथा β के मध्य समान्तर माध्य है, तो

$$\sin 2\theta + \cos 2\theta = 1 + \frac{nb(a-b)}{a^2+b^2}$$

जहाँ n कोई पूर्णांक है, तो निम्न प्रश्नों के उत्तर दीजिए।

28. The value of n is :

n का मान है :

(A) 0 (B) 1 (C) 2 (D) –2

Ans. C

Sol. Given $\cos\alpha + \cos\beta = a$, $\sin\alpha + \sin\beta = b$ & $\frac{\alpha+\beta}{2} = \theta$

$$\Rightarrow 2\cos\left(\frac{\alpha+\beta}{2}\right)\cos\left(\frac{\alpha-\beta}{2}\right) = a \quad ...(i)$$

$$2\sin\left(\frac{\alpha+\beta}{2}\right)\cos\left(\frac{\alpha-\beta}{2}\right) = b \quad ...(ii)$$

By (i) ÷ (ii); we get

$$\frac{a}{b} = \cot\theta \Rightarrow \tan\theta = \frac{b}{a}$$

$$\Rightarrow \sin\theta = \frac{b}{\sqrt{a^2+b^2}}, \cos\theta = \frac{a}{\sqrt{a^2+b^2}}$$

$\sin 2\theta + \cos 2\theta = 2\sin\theta\cos\theta + 2\cos^2\theta - 1$

$$= \frac{2ab}{a^2+b^2} + \frac{2a^2}{a^2+b^2} - 1$$

$$= \frac{a^2-b^2+2ab}{a^2+b^2} = 1 + \frac{2ab-2b^2}{a^2+b^2}$$

$$= 1 + \frac{2b(a-b)}{a^2+b^2}$$

$\Rightarrow n = 2$

29. If for n obtained in above question, $\sin^n A = x$, then $\sin A \sin 2A \sin 3A \sin 4A$ is a polynomial in x, of degree :
यदि ऊपर दिये गये प्रश्न में निकाले गये n के मान के लिए $\sin^n A = x$ हो, तो $\sin A \sin 2A \sin 3A \sin 4A$, x में एक बहुपद है जिसकी घात है :
(A) 5 (B) 6 (C) 7 (D) 8

Ans. A

Sol. Now $\sin^2 A = x$
$\sin A \sin 2A \sin 3A \sin 4A$
$= \sin A \cdot (3\sin A - 4\sin^3 A) \cdot \sin^2 2A \cdot \cos 2A$
$= (3\sin^2 A - 4\sin^4 A) \cdot 2 \cdot 4 \cdot \sin^2 A \cdot \cos^2 A \cdot (-2\sin^2 A + 1)$
$= (3x - 4x^2) \cdot 8 \cdot x \cdot (1-x)(1-2x)$
$= 8(3x - 4x^2) \cdot (x - x^2)(1-2x)$
$= 8(3x - 4x^2) \cdot (x - 3x^2 + 2x^3)$
degree is 5.

30. If degree of polynomial obtained in previous questins is p and $(p-5) + \sin x, \cos x, \tan x$ are in G.P., then $\cos^9 x + \cos^6 x + 3\cos^5 x - 1 =$
(A) –1 (B) 0 (C) 1 (D) none of these
यदि पहले वाले प्रश्न में निकाली गई बहुपद की घात p है तथा $(p-5) + \sin\theta, \cos\theta, \tan\theta$ गुणोत्तर श्रेढ़ी में है, तो $\cos^9 x + \cos^6 x + 3\cos^5 x - 1 =$
(A) –1 (B) 0 (C) 1 (D) इनमें से कोई नहीं

Ans. B

Sol. P = 5
$0 + \sin x, \cos x, \tan x \rightarrow$ G.P.

$$\Rightarrow \cos^2 x = \frac{\sin^2 x}{\cos x}$$

$\Rightarrow \cos^3 x = \sin^2 x$
$\Rightarrow \cos^9 x + \cos^6 x + 3\cos^2 x \cdot \cos^3 x - 1$
$= (\sin^2 x)^3 + \cos^6 x + 3\sin^2 x \cos^2 x - 1$
$= \sin^6 x + \cos^6 x + 3\sin^2 x \cos^2 x - 1$
$= (\sin^2 x + \cos^2 x)^3 - 3\sin^2 x \cos^2 x (\sin^2 x + \cos^2 x) + 3\sin^2 x \cos^2 x - 1$
$= 0$

Paragraph for Question No. 31 to 33

Let p be the product of the sines of the angles of a triangle ABC and q is the product of the cosines of the angles.

मानाकि एक त्रिभुज ABC के कोणों की ज्याओं (sines) का गुणनफल p है और कोणों की कोज्याओं (cosines) का गुणनफल q है।

31. In this triangle $\tan A + \tan B + \tan C$ is equal to :

(A) $p + q$ (B) $p - q$ (C) $\frac{p}{q}$ (D) none of these

इस त्रिभुज में $\tan A + \tan B + \tan C$ =

(A) $p + q$ (B) $p - q$ (C) $\frac{p}{q}$ (D) इनमें से कोई नहीं

Ans. C

Sol. Given $\sin A \sin B \sin C = P$ $\cos A \cos B \cos C = q$

in ΔABC

$A + B + C = \pi$

$\Rightarrow \tan A + \tan B + \tan C = \tan A \cdot \tan B \cdot \tan C$

$$= \frac{\sin A \cdot \sin B \cdot \sin C}{\cos A \cos B \cos C}$$

$$= \frac{p}{q}$$

32. $\tan A \tan B + \tan B \tan C + \tan C \tan A =$

$\tan A \tan B + \tan B \tan C + \tan C \tan A =$

(A) $1 + q$ (B) $\frac{1+q}{q}$ (C) $1 + p$ (D) $\frac{1+p}{p}$

Ans. B

Sol. $\tan A \tan B + \tan B \tan C + \tan C \tan A$

$$= \frac{\sin A \sin B}{\cos A \cos B} + \frac{\sin B \sin C}{\cos B \cos C} + \frac{\sin A \sin C}{\cos A \cos C}$$

$$= \frac{\cos C \sin A \sin B + \cos A \sin B \sin C + \sin A \cos B \sin C}{\cos A \cos B \cos C}$$

$$= \frac{\sin B \cdot (\sin A \cos C + \cos A \sin C) + \sin A \cos B \sin C}{q}$$

$$= \frac{1}{q}\left[\sin B \cdot \sin(A+C) + \sin A \cdot \cos B \cdot \sin C\right]$$

$$= \frac{1}{q}\left[\sin^2 B + \sin A \cos B \cdot \sin C\right]$$

$$= \frac{1}{q}\left[1 - \cos^2 B + \sin A \cos B \sin C\right]$$

$$= \frac{1}{q}\left[1 - \cos B\left(\cos B - \sin A \cdot \sin C\right)\right]$$

$$= \frac{1}{q}\left[1 - \cos B\left\{-\cos(A+C) - \sin A \sin C\right\}\right]$$

$$= \frac{1}{q}\left[1 + \cos B \cdot \cos A \cdot \cos C\right] = \frac{1+q}{q}$$

33. The value of $\tan^3 A + \tan^3 B + \tan^3 C$ is :

$\tan^3 A + \tan^3 B + \tan^3 C$ का मान है :

(A) $\frac{p^3 - 3pq^2}{q^3}$ (B) $\frac{q^3}{p^3}$ (C) $\frac{p^3}{q^3}$ (D) $\frac{p^3 - 3pq}{q^3}$

Ans. D

Sol. $(\tan A + \tan B + \tan C)^3$

$= \tan^3 A + \tan^3 B + \tan^3 C + 3(\tan A + \tan B + \tan C) \cdot (\tan A \cdot \tan B + \tan B \tan C + \tan C \tan A) - 3\tan A \tan B \tan C$

$$\Rightarrow \frac{p^3}{q^3} = \tan^3 A + \tan^3 B + \tan^3 C + 3\left(\frac{1+q}{q}\right)\cdot\frac{p}{q} - 3\frac{p}{q}$$

$$\Rightarrow \frac{p^3}{q^3} - \frac{3p}{q^2} = \tan^3 A + \tan^3 B + \tan^3 C$$

$$\Rightarrow \tan^3 A + \tan^3 B + \tan^3 C = \frac{p^3 - 3pq}{q^3}$$

MATCH THE COLUMN TYPE QUESTION(S)

34. Match the following column :

Column I	**Column II**
(A) The value of $\cos\frac{\pi}{19}+\cos\frac{3\pi}{19}+\cos\frac{5\pi}{19}+.....+\cos\frac{17\pi}{19}$ is	(P) – 3
(B) Exact value of $\cos^4\frac{\pi}{16}+\cos^4\frac{3\pi}{16}+\cos^4\frac{5\pi}{16}+\cos^4\frac{7\pi}{16}$ is	(Q) $\frac{3}{16}$
(C) The value of $6(\sin^6\theta+\cos^6\theta)-9(\sin^4\theta+\cos^4\theta)$ is	(R) $\frac{1}{2}$
(D) The value of the expression $\sin 20°\sin 40°\sin 60°\sin 100°$ is	(S) $\frac{3}{2}$

निम्न कॉलम को सुमेलित कीजिए :

Column I	**Column II**
(A) $\cos\frac{\pi}{19}+\cos\frac{3\pi}{19}+\cos\frac{5\pi}{19}+.....+\cos\frac{17\pi}{19}$ का मान होगा	(P) – 3
(B) $\cos^4\frac{\pi}{16}+\cos^4\frac{3\pi}{16}+\cos^4\frac{5\pi}{16}+\cos^4\frac{7\pi}{16}$ का मान होगा	(Q) $\frac{3}{16}$
(C) $6(\sin^6\theta+\cos^6\theta)-9(\sin^4\theta+\cos^4\theta)$ का मान होगा	(R) $\frac{1}{2}$
(D) $\sin 20°\sin 40°\sin 60°\sin 100°$ का मान होगा	(S) $\frac{3}{2}$

Ans (A) R; (B) S; (C) P; (D) Q

Sol. (A) $\cos\left(\frac{\pi}{19}\right)+\cos\left(\frac{3\pi}{19}\right)+.....+\cos\left(\frac{17\pi}{19}\right)$

$=\cos\left(\frac{\pi}{19}\right)+\cos\left(\frac{\pi}{19}+\frac{2\pi}{19}\right)+\cos\left(\frac{\pi}{19}+\frac{4\pi}{19}\right)+.....+\cos\left(\frac{\pi}{19}+\frac{16\pi}{19}\right)$

$=\frac{\sin\left(\frac{9}{2}\cdot\frac{2\pi}{19}\right)}{\sin\left(\frac{\pi}{19}\right)}\cos\frac{\left(\frac{2\pi}{19}+\frac{16\pi}{19}\right)}{2}$

$=\frac{\sin\left(\frac{9\pi}{19}\right)\cos\left(\frac{9\pi}{19}\right)}{\sin\left(\frac{\pi}{19}\right)}=\frac{\sin\left(\frac{18\pi}{19}\right)}{2\sin\left(\frac{\pi}{19}\right)}=\frac{1}{2}$

(B) $\cos^4\frac{\pi}{16}+\cos^4\frac{3\pi}{16}+\cos^4\frac{5\pi}{16}+\cos^4\frac{7\pi}{16}$

$=\cos^4\frac{\pi}{16}+\cos^4\frac{5\pi}{16}+\cos^4\left(\frac{\pi}{2}-\frac{5\pi}{16}\right)+\cos^4\left(\frac{\pi}{2}-\frac{7\pi}{16}\right)$

$=\cos^4\frac{\pi}{16}+\sin^4\frac{\pi}{16}+\cos^4\frac{5\pi}{16}+\sin^4\frac{5\pi}{16}$

$=\left(\sin^2\frac{\pi}{16}+\cos^2\frac{\pi}{16}\right)^2-2\sin^2\frac{\pi}{16}\cos^2\frac{\pi}{16}+\left(\sin^2\frac{5\pi}{16}+\cos^2\frac{5\pi}{16}\right)^2-2\sin^2\frac{5\pi}{16}\cos^2\frac{5\pi}{16}$

$=1+1-\frac{1}{2}\left[\sin^2\frac{\pi}{8}+\sin^2\frac{5\pi}{8}\right]$

$=2-\frac{1}{2}\left[\sin^2\frac{\pi}{8}+\sin^2\left(\frac{\pi}{2}+\frac{\pi}{8}\right)\right]$

$=2-\frac{1}{2}=\frac{3}{2}$

(C) $6\,[\sin^6\theta+\cos^6\theta]-9\,(\sin^4\theta+\cos^4\theta)$

$=6\,[(\sin^2\theta+\cos^2\theta)^3-3\sin^2\theta\cos^2\theta]-9\,((\sin^2\theta+\cos^2\theta)^2-2\sin^2\theta\cos^2\theta)$

$=6-18\sin^2\theta\cos^2\theta-9+18\sin^2\theta\cos^2\theta$

$=-3$

(D) $\sin 20^\circ\sin 40^\circ\sin 60^\circ\sin 100^\circ$

$=\frac{\sqrt{3}}{2}\cdot\sin 40^\circ\sin\left(60^\circ+40^\circ\right)\sin\left(60^\circ-40^\circ\right)$

$=\frac{\sqrt{3}}{2}\cdot\frac{1}{4}\cdot\sin\left(120^\circ\right)=\frac{1}{4}\cdot\frac{\sqrt{3}}{2}\cdot\frac{\sqrt{3}}{2}$

$=\frac{3}{16}$